Stoma Companion — Mesh Processing & Outline Capture

Contractor reference for the macOS Companion stack: USDZ intake through scaled perimeter export. Scope is CompanionMac + SharedPhotogrammetry + firmware as path consumer.

macOS StomaCompanion | SharedPhotogrammetry | Firmware = path executor

macOS 14+
Companion platform

100
Outline samples

DICT_4X4_50
ArUco dictionary

USDZ
Mesh interchange

Input is a textured USDZ (imported or reconstructed on Mac). Companion slices a closed base perimeter, applies real-world scale, and exports Cartesian G-code and/or polar jobs. ArUco supplies scale and marker exclusion; it is not used as the flange slice plane.

In scope

Component	Responsibility
Stoma Companion (macOS)	USDZ intake, optional Object Capture, perimeter slice, scale, path export
SharedPhotogrammetry	RealityKit PhotogrammetrySession + video frame export used by Companion
Firmware / host tools	Consumes exported .gcode / .polar; does not participate in meshing or metrology

Pipeline

Stage	I/O	Modules
Intake	Image folder, video, or existing USDZ	CompanionContentView, FolderImageImporter, VideoFrameExporter
Reconstruct (optional)	Image folder → model.usdz	PhotogrammetryProcessor (PhotogrammetrySession)
Scale	USDZ (± calibration still) → userScale + marker exclusion	MeshArUcoOrbitDetector, ArUcoDetectorBridge, PhotoScaleEstimator, ReferenceObjectMeasure
Outline	USDZ → closed 100-sample perimeter in slice plane	BasePerimeterExtractor
Export	Samples → Cartesian .gcode and/or polar .polar (+ Ideal Fit)	BasePerimeterExport, PolarPathExport, IdealFitOutlineGenerator, PlotterGcodeAutoExport
Execute	Path files → plotter motion	send_gcode.py / send_polar.py → StomaPlotter.ino

Responsibility layers

SwiftUI / policy

Companion views choose scale source, slice parameters, and export mode. They own workflow state and metrology decisions.

Geometry (Swift)

BasePerimeterExtractor and exporters are pure/static: no UI, ModelIO + simd only.

Computer vision (ObjC++)

ArUcoDetectorBridge answers marker location (and photo homography). Swift does not implement the detector.

Firmware

Executes exported paths only. No mesh load, slice, or scale logic on device.

Modules

Module	Path	Role	Structure
CompanionContentView	CompanionMac/	App shell: optional reconstruction and USDZ handoff into the perimeter tool	SwiftUI; embeds BasePerimeterToolView; FolderImageImporter for Object Capture input layout
PhotogrammetryProcessor	SharedPhotogrammetry/	Object Capture reconstruction to model.usdz when images/video are supplied on Mac	@MainActor ObservableObject wrapping PhotogrammetrySession; progress and cancel
VideoFrameExporter	SharedPhotogrammetry/	Samples video to JPEG frames; writes calibration_top.jpg at t≈0 for photo scale	AVAssetImageGenerator; shared defaults for interval and frame cap
BasePerimeterToolView	CompanionMac/	Operator UI for slice orientation, scale source, extract, preview, export, validation	SwiftUI orchestration; ScaleCalibrationSource enum; delegates geometry and CV to helpers
BasePerimeterExtractor	CompanionMac/	Loads USDZ, resolves floor/slice frame, intersects mesh, traces closed loop, resamples outline	Static enum API; ModelIO triangles; FloorDetectionResult / BasePerimeterResult types
MeshArUcoOrbitDetector	CompanionMac/	Virtual-camera orbit of textured USDZ; ArUco corners → scene scale and exclusion AABB	SceneKit offscreen renders; unproject + reprojection-selected screen-axis mapping
ArUcoDetectorBridge	CompanionMac/ (.h / .mm)	DICT_4X4_50 marker detection and optional pixel→mm homography	ObjC++ implementation; Swift via bridging header; marker generator for printable targets
Photo scale path	CompanionMac/	Top-down still ↔ mesh outline scale (ArUco, coin lines, or multi-metric estimate)	PhotoStomaContourDetector, StomaShapeMetrics, PhotoScaleEstimator, QuarterPhotoCalibrationView
ReferenceObjectMeasure	CompanionMac/	Scale from a named USDZ subtree AABB with known physical size	ModelIO name walk → extent → scale factor
Export & Ideal Fit	CompanionMac/	Cartesian G1 ring, polar M200–M202 job, clearance-offset Ideal Fit path	Pure builders from BasePerimeterResult; auto-write under firmware/test_patterns/
Previews & validation	CompanionMac/	3D slice preview, 2D path preview, operator validation chords and report export	SceneKit / Canvas views; OutlineValidationLineEditor; ValidationExport

BasePerimeterExtractor

Core outline algorithm — CompanionMac/BasePerimeterExtractor.swift

Step	Detail
Load	MDLAsset → world-space triangles (optional name / AABB exclude for marker geometry)
Frame	resolveSliceFrame: automatic support plane or fixed / manual tilt axes
Intersect	Triangle–plane segments in slice (U, V) coordinates
Loop	Snap segment graph → largest closed perimeter polygon
Sample	Arc-length resample to 100 points (θ, r, x, y) → BasePerimeterResult

Scale sources

Source	Behavior
ArUco in USDZ	Virtual-camera detect on mesh texture; known marker side (mm) → userScale; AABB exclusion
Photo + ArUco / metrics	Calibration still; contour and/or marker; multi-metric match to mesh outline
Coin / 1D reference	Interactive lines or catalog sizes on calibration still
Named mesh object	Known real size of a named USDZ subtree
Manual	Operator-entered scale factor

Mesh ArUco (virtual camera)

Orbit

Offscreen SceneKit renders of the textured USDZ from multiple viewpoints (~28).

Detect

ArUcoDetectorBridge on each frame; select a view that passes geometry checks.

Unproject

Raycast corners to the mesh; set userScale from known marker side (mm) and mean scene side; build WorldAABBExclusion for the marker.

Key types

Type	Meaning
BasePerimeterResult	Closed outline samples plus slice-plane frame for export
BasePlaneSample	One (θ, r, x, y) sample in the slice plane
WorldAABBExclusion	Axis-aligned box used to drop marker triangles from the subject
MeshArUcoOrbitResult	World corners, mean side in scene units, confidence, optional debug PNG
ArUcoHomographyResult	3×3 row-major pixel→mm transform for photo workflows
ManualSliceAxisTuning	Base axis + Euler tilt + in-plane spin for the slice frame
PolarPathPlan	Fixed-ω one-revolution lead-screw schedule (M200–M202)

Units & outputs

Scene units are scaled by userScale. Millimeter export (G21) multiplies by 1000. Cartesian export uses G28 / G21 / G90 / G1 / M2. Polar export uses stoma_polar_v1 with M200 (RPM), M201 (segment table), M202 (one CCW revolution). Ideal Fit is an outward-offset clearance ring derived from the same outline.

Code root: ios/StomaScanner/CompanionMac and SharedPhotogrammetry · Target: StomaCompanion